Mikhwah General Hospital
AL-BAHA CLUSTER

DPP	**DEPARTMENT POLICY AND PROCEDURE**		
Title	**USING OF PULSE OXIMETRY**		
Department	**Intensive Care Unit**		
Applies to	**All ICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-ICU-033 (v3)**		
Last Version Approval Date	**15/12/2022**	Number of Pages	3

	NAME	POSITION	SIGNATURE	DATE
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Reviewed by:	**DR. SALAH MOHD MANSOUR**	Head of Department, Intensive Care Unit		13-11-25
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Approved by:	**MR. ABDULMAJEED MOHD AL-SOHEIMI**	Nursing Director		18.11.25
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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
18 / 11 / 2025	2 / 12 / 2025	TRIENNIAL	17 / 11 / 2028

1.0 PURPOSE

1.1. To continuously monitor oxygen saturation of arterial blood.

1.2. To assess patient's response to oxygen therapy.

1.3. To evaluate the need for oxygen therapy.

2.0 DEFINITION/ABBREVIATION

2.1. **Pulse Oximetry**- is a non-invasive saturation monitoring where light sensor is taped to a limb or the ear. It is useful in determining the oxyhemoglobin saturation of patient at risk for hypoxia

3.0 RESPONSIBILITY

3.1 **Staff nurse** – observation and recording of patient's oxygen saturation whether with ongoing oxygen therapy or on room air.

4.0 POLICY

4.1. Calibration must be done by the Bio-medical technician.

4.2. Pulse oximeter probe must be cleaned with alcohol before and after each patient's use.

4.3. Attachment of the probe is done by qualified nursing personnel.

5.0 PROCEDURE (S)

5.1. Wash hand.

5.2. Assemble equipments.

5.3. Explain the procedure to the patient and/or to his family.

5.4. Select adequate site for application of the probe (lateral side of palm).

5.5. Use the proper equipment. Probe should be appropriate to patient's size and age.

5.6. Prepare the monitoring site by cleaning the selected area and allowing it to dry.

5.7. Apply the probe securely to the skin. Make sure that the light-emitting sensor and the light receiving sensor are aligned opposite each other.

5.8. Connect the probe to the pulse oximeter and check the operation of the equipment.

5.9. Always position the patient's hand at heart level.

5.10. Set alarm on the pulse oximeter.

5.11. Check oxygen saturation at regular intervals as ordered by physician. Monitor patient's haemoglobin.

5.12. Remove sensor on a regular basis and check for skin irritation or signs of pressure.

5.13. Document the procedure including the date, time, procedure type, oximetric measurement and any action taken. Record reading on appropriate form.

6.0 EQUIPMENTS

6.1. Pulse oximeter

7.0 ATTACHMENT/FORMS

7.1. ICU Daily Flow Sheet

7.2. Vital signs chart (GDOH-NUR-VSC-193)

7.3. Nurses Progress Notes (NUR-NPN-203)

8.0 REFERENCES

8.1. Nettina, S. (8th.).(2006). The Lippincott manual of Nursing Practice. USA

8.2. Weigand DL,6th ed (2010): AACN procedure manual for critical care, Philadelphia

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

MIKHWA GENERAL HOSPITAL
AL-MIKHWA, AL- BAHA, KSA
INTENSIVE CARE UNIT

MRN ☐☐☐☐☐☐☐☐☐☐

Name:

Age: ☐☐ Yrs/Months Sex: M ☐ F ☐

Nationality:

Consultant In charge:

Dept: Unit:

DAILY FLOW SHEET

PRIMARY PHYSICIAN	CONSULTANT ON ADMISSION		SURGICAL PROCEDURES	
CONSULTANTS			(I)	DATE
1	HOSPITAL ADMISSION DATE	ICU ADMISSION DATE	(II))	DATE
2				
3			(III)	DATE
4				
5				
PATIENT HISTORY	ADMITTED FROM:		(III)	DATE
DIAGNOSIS	NO. OF DAYS IN ICU		(IV)	DATE
PRECAUTIONS	ALLERGIES		(V)	DATE

DATE	INVESTIGATIONS	COMMENTS	DATE	DIAGNOSTIC PROCEDURES	COMMENTS

NURSING CARE PLAN

DATE	TIME	PROBLEMS/ NURSING DIAGNOSIS	GOAL / EXPECTED OUTCOME	INTERVENTION	EVALUATION	NAME/ID NO./ SIGNATURE

Peak Temperature/ Daily Balance (Last 5 Days)		INTAKE	0700H	0800H	0900H	1000H	1100H	1200H	1300H	1400H	8 hours
Date	°C										
		IN-TOTAL									
Patient's Weight	Height	OUTPUT									
APACHE IV Score											
P.M.R.											
		OUT-TOTAL									
L.O.S		VENTILATOR MODE										BALANCE
		FiO_2									
		Machine Rate / Spon. Rate									
Diet / Tube Feeding		V_T									
		PEEP									
		Pressure Support									
		Blood Glucose / Insulin Given									TOTAL INSULIN

CODE			Resp	Pulse & BP	Temp	0700H	0800H	0900H	1000H	1100H	1200H	1300H	1400H
		BLOOD PRESSURE (Red)		220	43								
	(Resp. Rate – Black)			200	42								
●		˅ ARTERIAL		180	41								
		˄		160	40								
				140	39								
●	(Pulse – Red)	˄	60	120	38								
		CUFF	50	100	37								
		˅	40	80	36								
X			30	60	35								
	(Temp – blue)		20	40	34								
			10	20	33								

	0700H	0800H	0900H	1000H	1100H	1200H	1300H	1400H
SPO2								
SVO2								
CARDIAC RHYTHM								
MAP								
CVP								
CO / CI								
PWCP								
ICP								
ABDOMINAL ☐ / GIRTH ☐ / IAP ☐								

INTAKE	1500H	1600H	1700H	1800H	1900H	2000H	2100H	2200H	8 hours
IN-TOTAL									
OUTPUT									
OUT-TOTAL									

Diet / Tube Feeding										
	VENTILATOR MODE									BALANCE
	FiO_2									
	Machine Rate / Spon. Rate									
	V_T									
	PEEP									
	Pressure Support									
	Blood Glucose / Insulin Given									TOTAL INSULIN

CODE			Resp	Pulse & BP	Temp	1500H	1600H	1700H	1800H	1900H	2000H	2100H	2200H
	(Temp – blue) (Pulse – Red) (Resp. Rate – Black)			220	43								
●		BLOOD PRESSURE (Red)		200	42								
				180	41								
		˅		160	40								
		ARTERIAL		140	39								
●		˄	60	120	38								
			50	100	37								
		˄	40	80	36								
X		CUFF ˅	30	60	35								
			20	40	34								
			10	20	33								

SPO2								
SVO2								
CARDIAC RHYTHM								
MAP								
CVP								
CO / CI								
PWCP								
ICP								
ABDOMINAL ☐ / GIRTH ☐ / IAP ☐								

INTAKE	2300H	2400H	0100H	0200H	0300H	0400H	0500H	0600H	8 hours TOTAL	24 Hrs TOTAL
IN-TOTAL										
OUTPUT										
OUT-TOTAL									BALANCE	
VENTILATOR MODE									TOTAL INTAKE 24 Hrs.	
FiO_2										
Machine Rate / Spon. Rate									TOTAL OUTPUT 24 Hrs.	
V_T										
PEEP									TOTAL BALANCE	
Pressure Support										
Blood Glucose / Insulin Given									INSULIN 8 HRS.	INSULIN 24 HRS.

CODE			Resp	Pulse & BP	Temp	2300H	2400H	0100H	0200H	0300H	0400H	0500H	0600H
	(Temp – blue) (Pulse – Red) (Resp. Rate – Black)	BLOOD PRESSURE (Red)		220	43								
				200	42								
●				180	41								
		˅		160	40								
		ARTERIAL		140	39								
●		˄	60	120	38								
			50	100	37								
		˄	40	80	36								
X		CUFF	30	60	35								
		˅	20	40	34								
			10	20	33								

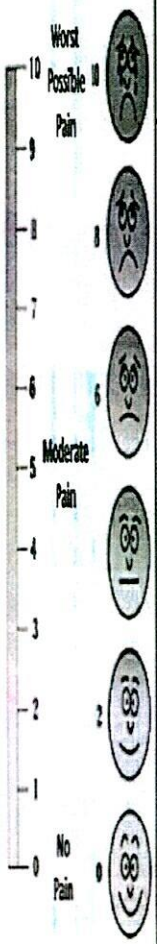
	2300H	2400H	0100H	0200H	0300H	0400H	0500H	0600H
SPO2								
SVO2								
CARDIAC RHYTHM								
MAP								
CVP								
CO / CI								
PWCP								
ICP								
ABDOMINAL ☐ / GIRTH ☐ / IAP ☐								

RAMSAY SEDATION SCALE
1. Anxious, agitated or restless
2. Cooperative, oriented and tranquil
3. Sedated but responds to light glabellar tap or loud auditory stimulus
4. Asleep; brisk responses to light glabellar tap or loud auditory stimulus
5. Asleep, sluggish responses to light glabellar tap or loud auditory stimulus
6. Asleep; no response to painful stimuli

Time (24 hour clock)				0700	0800	0900	1000	1100	1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	2400	0100	0200	0300	0400	0500	0600	
GLASGOW COMA SCALE	Eyes Open	Spontaneously	4																									Eyes closed by swelling = C
		To Speech	3																									
		To Pain	2																									
		None	1																									
	Verbal Response	Oriented	5																									Endo tracheal tube or Trache ostomy = T
		Confused	4																									
		Inappropriate Words	3																									
		Incomprehensive Sounds	2																									
		None	1																									
	Best Motor Response	Obeys Commands	6																									Usually record the best arm response
		Localize Pain	5																									
		Withdraws to Pain	4																									
		Flexion to pain	3																									
		Extension to pain	2																									
		None	1																									
RAMSAY SEDATION SCALE																												
PUPIL	Right	Size																										(+) reaction (-) no reaction (C) eye closed
		Reaction																										
	Left	Size																										
		Reaction																										
LIMB MOVEMENT	ARMS	Normal Power	6																									Record right (R) and Left (L) separately if there is a difference between the two sides
		Mild Weakness	5																									
		Severe Weakness	4																									
		Spastic Flexion	3																									
		Extension	2																									
		No Response	1																									
	LEGS	Normal Power	6																									
		Mild Weakness	5																									
		Severe Weakness	4																									
		Spastic Flexion	3																									
		Extension	2																									
		No Response	1																									
PAIN ASSESSMENT TOOL	PAIN SCORE		10																									***Reassessment of Pain must be completed within 15min to 1hr post:*** Oral Analgesics in 1 hr IM Analgesics in 30 mins SCut Analgesics in 20mins IV Analgesics in 15 mins SubL Analgesics in 15mins PR analgesics in 15 mins
			9																									
			8																									
			7																									
			6																									
			5																									
			4																									
			3																									
			2																									
			1																									
			0																									
	Location																											
	Quality																											
	Pharmaceutical																											
	Alternative Intervention																											
	Therapeutic Adverse Effect																											
	Reassessment Score																											

1
2
3
4
5
6
7
8
Pupil size in mm

Pain Rating Scales



	Parameters	Pain Assessment Score		
		0	1	2
NONVERBAL	Face	No Smile	Grimace	Tearing
	Activity	Neutral	Some movement	Restless
	Guarding	Neutral	Tensed	Rigid
	Physiology	Stable	Slight Change	Obvious Change
	Respiratory	Baseline	Slight Change	Severe Change

Therapeutic Adverse Effect	
R – Rash	D – Dizziness
V – Vomiting	C – Confusion
Co – Constipation	A – Allergic reaction
H – Hemodynamic Change	R – Respiratory Depression
N - Nausea	H – Hemodynamic Change

	Location				Quality		
VERBAL	1 – Head	4 – Chest	7 – UE or LE	10 - Flank	A - Aching	H - Heavy	S – Stabbing
	2 – Neck	5 - Abdomen	8 – Oper. Site	11 - Generalized	B - Burning	E - Electrical	Sh – Sharp
	3 – Shoulder	6 - Back	9 – Genital area		C - Cramping	R - Radiating	T - Throbbing

Date: MORNING

NEUROLOGICAL

I. Level of Consciousness
( ) Alert ( ) Disoriented
( ) Lethargic ( ) Coma ( ) Sedated

II. Reflex
( ) Gag ( ) Cough
( ) Babinski ( ) Corneal

III. Motor
( ) Normal ( ) Decorticate
( ) Decerebate ( ) Other

CARDIOVASCULAR

I. Rhythm: ( ) Regular ( ) Irregular
II. Heart Sound: ( ) S1/S2 ( ) Other
III. Edema: 0 1 2 3 4
Site: ____________
IV. Capillary Refill: <3 Sec >3 Sec
V. Extremity Coolness: ARM **R / L**
HAND **R / L** LEG **R / L** FOOT **R / L**
VI. Other :

RESPIRATORY

I. Breathing : ( ) Spontaneous
( ) Tracheostomy ( ) Ventilator ________
II. Cough:
( ) No Cough ( ) Productive
( ) Non Productive ( ) Weak ( ) Strong
III. Intubation: Date : **Size:**
Extubation: **Reintubation:**
IV. Secretions:
Amount: ____________
Consistency: ____________
V. Breath Sound: R/L Clear **R/L** Other
VI. Cyanosis: Location : ____________
VII. Chest Tube
Location: ________ Level: ________
Suction: ____________

GASTROINTESTINAL

I. LBM : Date: ________ Time : ________
II. Diet: ( ) Oral ( ) NGT ( ) OGT ( ) TPN
NPO Other: ____________
III. Abdomen: ( ) Soft ( ) Tender
( ) Flat ( ) Distended ( ) Ascites
IV. Bowel Sound:
( ) Present ( ) Absent
( ) Hypoactive ( ) Hyperactive
V. Drainage Tube: ____________

GENITO-URINARY

I. Urine: ( ) Clear ( ) Cloudy
Color: ____________
II. ( ) Void freely ( ) Incontinence
( ) Condom Cath ( ) Foley Catheter
III. Urinary Catheter Size: ________
Date inserted : ____________

INTEGUMENTARY

I. ( ) Intact ( ) Dry ( ) Moist ( ) Moist
( ) Cool
II. Bed Sores: ____________

III. Oral Membrane ( ) Intact ( ) Dry
Color : ____________

Date: EVENING

NEUROLOGICAL

I. Level of Consciousness
( ) Alert ( ) Disoriented
( ) Lethargic ( ) Coma ( ) Sedated

II. Reflex
( ) Gag ( ) Cough
( ) Babinski ( ) Corneal

III. Motor
( ) Normal ( ) Decorticate
( ) Decerebate ( ) Other

CARDIOVASCULAR

I. Rhythm: ( ) Regular ( ) Irregular
II. Heart Sound: ( ) S1/S2 ( ) Other
III. Edema: 0 1 2 3 4
Site: ____________
IV. Capillary Refill: <3 Sec >3 Sec
V. Extremity Coolness: ARM **R / L**
HAND **R / L** LEG **R / L** FOOT **R / L**
VI. Other :

RESPIRATORY

I. Breathing : ( ) Spontaneous
( ) Tracheostomy ( ) Ventilator ________
II. Cough:
( ) No Cough ( ) Productive
( ) Non Productive ( ) Weak ( ) Strong
III. Intubation: Date : **Size:**
Extubation: **Reintubation:**
IV. Secretions:
Amount: ____________
Consistency: ____________
V. Breath Sound: R/L Clear **R/L** Other
VI. Cyanosis: Location : ____________
VII. Chest Tube
Location: ________ Level: ________
Suction: ____________

GASTROINTESTINAL

I. LBM : Date: ________ Time : ________
II. Diet: ( ) Oral ( ) NGT ( ) OGT ( ) TPN
NPO Other: ____________
III. Abdomen: ( ) Soft ( ) Tender
( ) Flat ( ) Distended ( ) Ascites
IV. Bowel Sound:
( ) Present ( ) Absent
( ) Hypoactive ( ) Hyperactive
V. Drainage Tube: ____________

GENITO-URINARY

I. Urine: ( ) Clear ( ) Cloudy
Color: ____________
II. ( ) Void freely ( ) Incontinence
( ) Condom Cath ( ) Foley Catheter
III. Urinary Catheter Size: ________
Date inserted : ____________

INTEGUMENTARY

I. ( ) Intact ( ) Dry ( ) Moist ( ) Moist
( ) Cool
II. Bed Sores: ____________

III. Oral Membrane ( ) Intact ( ) Dry
Color : ____________

Date: NIGHT

NEUROLOGICAL

I. Level of Consciousness
( ) Alert ( ) Disoriented
( ) Lethargic ( ) Coma ( ) Sedated

II. Reflex
( ) Gag ( ) Cough
( ) Babinski ( ) Corneal

III. Motor
( ) Normal ( ) Decorticate
( ) Decerebate ( ) Other

CARDIOVASCULAR

I. Rhythm: ( ) Regular ( ) Irregular
II. Heart Sound: ( ) S1/S2 ( ) Other
III. Edema: 0 1 2 3 4
Site: ____________
IV. Capillary Refill: <3 Sec >3 Sec
V. Extremity Coolness: ARM **R / L**
HAND **R / L** LEG **R / L** FOOT **R / L**
VI. Other :

RESPIRATORY

I. Breathing : ( ) Spontaneous
( ) Tracheostomy ( ) Ventilator ________
II. Cough:
( ) No Cough ( ) Productive
( ) Non Productive ( ) Weak ( ) Strong
III. Intubation: Date : **Size:**
Extubation: **Reintubation:**
IV. Secretions:
Amount: ____________
Consistency: ____________
V. Breath Sound: R/L Clear **R/L** Other
VI. Cyanosis: Location : ____________
VII. Chest Tube
Location: ________ Level: ________
Suction: ____________

GASTROINTESTINAL

I. LBM : Date: ________ Time : ________
II. Diet: ( ) Oral ( ) NGT ( ) OGT ( ) TPN
NPO Other: ____________
III. Abdomen: ( ) Soft ( ) Tender
( ) Flat ( ) Distended ( ) Ascites
IV. Bowel Sound:
( ) Present ( ) Absent
( ) Hypoactive ( ) Hyperactive
V. Drainage Tube: ____________

GENITO-URINARY

I. Urine: ( ) Clear ( ) Cloudy
Color: ____________
II. ( ) Void freely ( ) Incontinence
( ) Condom Cath ( ) Foley Catheter
III. Urinary Catheter Size: ________
Date inserted : ____________

INTEGUMENTARY

I. ( ) Intact ( ) Dry ( ) Moist ( ) Moist
( ) Cool
II. Bed Sores: ____________

III. Oral Membrane ( ) Intact ( ) Dry
Color : ____________

		0700	0900	1100	1300	1500	1700	1900	2100	2300	0100	0300	0500
HYGIENE	**Bath: Complete Partial**												
	Mouth Care												
	Skin Care												
	Eye Care												
	Ear Care												
	Perineal / Foley Care												
	ETT/Trach Care												
ACTIVITY	Awake Sleeping Resting Sedated Coma												
	Ambulation: Self Assist Chair Dangle												
	Turned : Left Right Back Supine												
	Incentive Spirometer												
	Range of Motion (ROM) / Physio Therapist In												
	Suctioned: Oral Trach ET Nasopharynx												
ROUTINE CHECKS	**Restraint:** Wrist Ankle Vest												
	Reason: Agitated Confused *Other												
	Restrained Removed and Skin Assessed												
	NGT Placement Verified												
	Residual Amount												
	Bowel Motion: Soft Hard Loose Watery Melena												
	Splints: On Off												
	Elastic Stockings On Off Pneumatic Compressor												
	Bed Air Mattress: On Off												
	Thermo Regulator Cool Warm												

WOUND CONDITION

WOUND TYPE / LOCATION	APPEARANCE	DATE OF DRESSING	REMARKS

INVASIVE LINES AND TUBES

TYPE / LOCATION	FIXATION LEVEL	APPEARANCE	DATE OF INSERTION	DATE OF DRESSING	INSERTED BY	REMARKS

0700-1500	1500-2300	2300-0700	ITEMS CHECKED
☐	☐	☐	ID Band
☐	☐	☐	Ambubag
☐	☐	☐	Suction Set
☐	☐	☐	O_2 Flow Meter
☐	☐	☐	Equipment Alarms
☐	☐	☐	Cardiac Electrodes
☐	☐	☐	Stethoscope
☐	☐	☐	Side Rails Up
☐	☐	☐	Pulse Oximetry
☐	☐	☐	Lines Zeroed

SHIFT	0700-1500	1500-2300	2300-0700
STAFF NAME			
LEGIBLE SIGNATURE			

NURSES NOTES

Date	Time	Notes	Signature

NURSES NOTES

Date	Time	Notes	Signature

NURSES NOTES

Date	Time	Notes	Signature

BLOOD AND BLOOD PRODUCTS TRANSFUSION

Date/ Time	Blood Type	Quantity	Started At	Finished At	Any Reaction	Ordered by	Transfused by	Lab Result	
								Pre	Post

COMMENTS AND MEDICATIONS

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: ____________ :مستشفى

Region: ____________ :المنطقة/المحافظة

Dept./Unit: ____________ :القسم/الوحدة

MRN: ☐☐☐☐☐☐☐☐☐☐ :رقم الملف الطبي

Name: ____________ :الاسم

Nationality: ____________ :الجنسية

Age: ____________ ☐ سنة Years ☐ شهر Months ☐ يوم Days :العمر

Date of Birth: ____/____/14____ H ____/____/20____ :تاريخ الميلاد

Gender: ☐ Male ☐ Female :الجنس

VITAL SIGNS CHART

WEIGHT-kg						
POST-OPERATIVE DAY						
DATE						
TIME						
TEMPERATURE	**40**					
	39					
	38					
	37					
	36					
	35					
RESPIRATORY RATE						
O_2 SATURATION						
BLOOD PRESSURE AND PULSE RATE **PULSE = X** **SYSTOLIC = V** **DIASTOLIC = ^**	**240**					
	220					
	200					
	180					
	160					
	140					
	120					
	100					
	80					
	60					
	40					
	20					
BP						
PAIN SCORE SCALE						
INITIALS AND JOB NUMBER						

0 000000 001939

SN ☐

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: ______________ :مستشفى
Region: ______________ :المنطقة/المحافظة
Dept./Unit: ______________ :القسم/الوحدة

MRN: | | | | | | | | | | :رقم الملف الطبي
Name: ______________________________ :الاسم
Nationality: ______________________________ :الجنسية
Age: ______________ سنة ☐ Years شهر ☐ Months يوم ☐ Days :العمر
Date of Birth: _____ /_____ / 14_____ H _____ /_____ / 20_____ :تاريخ الميلاد
Gender: ☐ Male ☐ Female :الجنس

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

0 000000 002035

SN